

Lecture 1

Signal Energy

- Motivation: How to quantify the size/strength of a signal?
- What is the issue with $\int_{-\infty}^{\infty} x(t) dt$?
- Definition (only for **real signals**):

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt$$

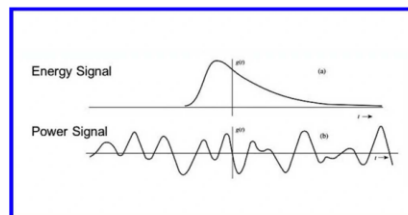
- When $x(t)$ is complex, $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$
- Example: $x(t) = \begin{cases} 2t, & -1 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$
- $x(t)$ is called an *energy signal* if $E_x < \infty$

Signal Power

- Motivation: We need another measure in case $E_x \rightarrow \infty$
- Definition:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$$

- $P_x \rightarrow \infty$ is still possible (e.g., $x(t) = t$)



- $x(t)$ is called a *power signal* if $0 < P_x < \infty$

Signal power 不一定要是周期函数，非周期函数也可以算 signal power

Lecture 2

Basic System Properties

- Static and Dynamic Systems
- Causal and Non-Causal Systems
- Invertible and Non-Invertible Systems
- Stable and Unstable Systems
- Time-Invariant and Time-Variant Systems
- Linear and Non-Linear Systems

系统类别判定总表

系统类别	核心问题	判定方法	一句话标准	常见误区
Static / Dynamic	当前输出是否只依赖当前输入？	看 $y(t)$ 或 $y[n]$ 是否只含同一时刻的 $x(t)$ / $x[n]$	若输出只由当前时刻输入决定，则 static (memoryless)；只要涉及过去/未来输入、积分、微分、延时、差分、累加，就是 dynamic	很多人以为“没有积分微分就是 static”，错。只要有 $x(t-1)$ 、 $x[n+1]$ 这种别的时刻，也不是 static
Causal / Non-causal	当前输出会不会用到未来输入？	只检查 $y(t_0)$ 或 $y[n_0]$ 是否依赖 $x(t > t_0)$ 或 $x[n > n_0]$	只依赖当前和过去输入就是 causal；只要依赖未来输入就是 non-causal	动态不等于非因果。 $y[n] = x[n-1]$ 是 dynamic，但仍然 causal
Invertible / Non-invertible	能不能由输出唯一恢复输入？	问：不同输入会不会得到同一个输出？若会，则不可逆	若一一对应，则 invertible；若多对一，则 non-invertible，看有没有信息的丢失	$x(t)$
Stable / Unstable	有界输入会不会产生无界输出？	用 BIBO stability：若对任意有界输入，输出都有限界，则稳定	输入有界 $\Rightarrow$ 输出有界，则 stable	别拿“零输入响应会不会发散”直接替代 BIBO。 Signals 基础里通常先按 bounded input $\rightarrow$ bounded output 判
Time-invariant / Time-variant	输入整体延迟后，输出会不会同样延迟？	做延时实验：先求 $T\{x(t-t_0)\}$ ，再比较是否等于 $y(t-t_0)$	若 $T\{x(t-t_0)\} = y(t-t_0)$ ，则 time-invariant	最大雷区：系统里出现 t 或 n 不一定就时变，但大概率要警惕；必须按定义验，不要靠感觉
Linear / Nonlinear	是否满足叠加原理？	检查 $T\{ax_1 + bx_2\} = ? aT\{x_1\} + bT\{x_2\}$	满足加法性和齐次性，就是 linear	$x(t)$

$$y(t) = \frac{d}{dt}x(t)$$

是 causal system。

符号	含义	定义/说明
$\text{Re}[x]$	实部	$\frac{x(t) + x^*(t)}{2}$
$\text{Im}[x]$ 或 $\text{imag}[x]$	虚部	$\frac{x(t) - x^*(t)}{2j}$
$\text{Ev}[x(t)]$	偶部	$\frac{x(t) + x(-t)}{2}$

符号	含义	定义/说明
$\text{Od}[x(t)]$	奇部	$\frac{x(t) - x(-t)}{2}$

Time-invariant and Time-variant Systems

1. $y(t) = x(2t)$
2. $y(t) = x(t+2)$
3. $y(t) = x(\sin t)$
4. $y(t) = t^2 x(t)$
5. $y(t) = 5x(t-1)$
6. $y(t) = \cos t + x(t)$
7. $y(t) = \frac{d}{dt} x(t)$
8. $y(t) = x(t) + 2$
9. $y(t) = e^{-3x(t)}$

Observation: For time-invariant systems:

- No time scaling, reversal, or other functionalities on t inside $x(\cdot)$
- Coefficient of $x(\cdot)$ should be independent of t (being constant)
- Any added term alongside $x(\cdot)$ must be constant or zero

1. 系统延时：把所有的 t 都换成 $t-t_0$

原输出：

$$y(t) = x(2t)$$

延时 t_0 后：

$$y(t - t_0) = x(2t - 2t_0)$$

2. 输入延时： $x(t)$ 变成 $x(t-t_0)$

延时输入：

$$x_1(t) = x(t - t_0)$$

$$y_1(t) = x(2t - t_0)$$

Linear and Non-linear Systems

1. $y(t) = x(t^2)$
2. $y(t) = t^2 x(t)$
3. $y(t) = x(t) + t^2$
4. $y(t) = \log[x(t)]$
5. $y(t) = x(\log t)$
6. $y(t) = \sin t \cdot x(t)$
7. $y(t) = \frac{d}{dt} x(t)$
8. $y(t) = \text{Even}[x(t)]$
9. $y(t) = \text{Re}[x(t)]$

Observation: For linear systems:

- No non-linear functionalities on $x(\cdot)$
- Coefficients of $x(\cdot)$ and t could be anything
- Any added term along side $x(\cdot)$ must be zero (zero input zero output)

设系统为：

$$y(t) = T\{x(t)\}$$

路径 A: 先合成输入, 再进系统

把输入换成:

$$x(t) = ax_1(t) + bx_2(t)$$

得到:

$$T\{ax_1(t) + bx_2(t)\}$$

路径 B: 分别进系统, 再线性组合输出

先算:

$$T\{x_1(t)\}$$

$$T\{x_2(t)\}$$

然后组合:

$$aT\{x_1(t)\} + bT\{x_2(t)\}$$

Lecture 3

Importance of Impulse Response

- For an LTI system, $h(t)$ fully determines the system's behavior
- The output of an LTI system for any given $x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)d\tau$ is

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

- $h(t)$ plays a key role in analyzing system properties such as stability and causality
- An intuitive explanation of system output

Determining Causality through $h(t)$

A system is **causal** if its output at any time t depends only on present and past inputs, but not on future inputs. This means the impulse response $h(t)$ satisfies:

$$h(t) = 0, \quad \forall t < 0$$

To see this, we have $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau$, which can be broken into two parts

- Example of a causal system:

$$h(t) = u(t)$$

- Example of a non-causal system:

$$h(t) = e^{|t|}$$

Determining Stability through $h(t)$

A system is **stable** if and only if its impulse response $h(t)$ satisfies the **bounded-input bounded-output (BIBO) stability criterion**:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

- Example of a stable system:

$$h(t) = e^{-t}u(t)$$

- Example of an unstable system:

$$h(t) = e^t u(t)$$

假设输入 $x(t)$ 有界:

$$|x(t)| \leq M_x$$

输出为:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

对两边取绝对值:

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \right|$$

所以:

$$|y(t)| \leq \int_{-\infty}^{\infty} |x(\tau)| |h(t-\tau)| d\tau$$

因为:

$$|x(\tau)| \leq M_x$$

所以:

$$|y(t)| \leq M_x \int_{-\infty}^{\infty} |h(t-\tau)| d\tau$$

如果:

$$\int_{-\infty}^{\infty} |h(\lambda)| d\lambda = M_h < \infty$$

那么:

$$|y(t)| \leq M_x M_h < \infty$$

所以输出有界, 系统稳定。

Common Properties of Convolution

The convolution operation satisfies the following key properties:

- **Commutativity:** $x_1(t) * x_2(t) = x_2(t) * x_1(t)$
- **Associativity:** $(x_1 * x_2) * x_3 = x_1 * (x_2 * x_3)$
- **Distributivity:** $x_1 * (x_2 + x_3) = x_1 * x_2 + x_1 * x_3$
- **Shift Property:** If $x_1(t) * x_2(t) = y(t)$, then

$$x_1(t) * x_2(t - T) = x_1(t - T) * x_2(t) = y(t - T)$$

- **Identity:** $x(t) * \delta(t) = x(t)$

Linear Constant-Coefficient Differential Equations (LCCDEs)

- A linear constant-coefficient differential equation is of the form:

$$a_n \frac{d^n}{dt^n} y(t) + \cdots + a_1 \frac{d}{dt} y(t) + a_0 y(t) = b_m \frac{d^m}{dt^m} x(t) + \cdots + b_1 \frac{d}{dt} x(t) + b_0 x(t)$$

- The coefficients a_i and b_i are constants, making LCCDEs **LTI systems**
- For simplicity, we can write $\frac{d}{dt} x(t) \rightarrow \dot{x}(t)$, $\frac{d^2}{dt^2} y(t) \rightarrow \ddot{y}(t)$ and so on
- Applications: Electrical circuits, mechanical systems, control systems, etc.

Calculating Impulse Response of LCCDEs

Step 1: Find the homogeneous solution $y_h(t)$

Step 2: Multiply $y_h(t)$ with $u(t)$ to make it causal

Step 3: Add singular functions to $y(t)$ if necessary

Step 4: Compute the higher-order derivatives of $y(t)$

Step 5: Determine the coefficients by matching singular terms on both sides

Exercise: Find $y(t)$ when $x(t) = \delta(t)$

- $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = \dot{x}(t) + 3x(t)$ ($a = 2, b = -1$)
- $\dot{y}(t) + 2y(t) = \ddot{x}(t) + 3\dot{x}(t) + 3x(t)$ ($a = b = c = 1$)

Signal Decomposition and Eigenfunction e^{st}

- For an LTI system with $h(t)$, $T[e^{st}] = H(s) \cdot e^{st}$
 - e^{st} is called an eigenfunction of the system
 - $H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$ is called the frequency response of $T[\cdot]$
- For a general signal $x(t) = a_1e^{s_1t} + a_2e^{s_2t}$, $T[x(t)] = a_1H(s_1)e^{s_1t} + a_2H(s_2)e^{s_2t}$
- e^{s_1t}, e^{s_2t} have to satisfy: $\langle e^{s_1t}, e^{s_1t} \rangle = 1$, $\langle e^{s_2t}, e^{s_2t} \rangle = 1$, $\langle e^{s_1t}, e^{s_2t} \rangle = 0$

Observation: An LTI system $T[\cdot]$ may magnify or eliminate certain frequencies, but will never create new frequencies

<, > 是求内积

Fourier Series Expansion of Periodic Signals

- We consider **periodic signal** $x(t)$ with fundamental period T ($\omega_0 = \frac{2\pi}{T}$)
- Eigenfunctions are defined as $e^{jk\omega_0t}$, $k = 0, \pm 1, \pm 2, \dots$
- Inner product is defined as $\langle f(t), g(t) \rangle = \frac{1}{T} \int_T f(t)[g(t)]^* dt$
- Fourier series expansion:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0t}$$

- Fourier series coefficients:

$$a_k = \langle x(t), e^{jk\omega_0t} \rangle = \frac{1}{T} \int_T x(t) e^{-jk\omega_0t} dt$$

信号内积的定义中

需要使用共轭，是为了与自身求内积的时候，要等于长度平方
使用积分是为了求所有的 $x_i y_i$ ，参考向量的内积的求法

Lecture 7

Symmetry for Real Signals

- **Conjugate symmetric property:**

$$x(t) = x^*(t) \iff a_k = a_{-k}^*$$

- Proof: two methods

- **Implications:**

1. $|a_k| = |a_{-k}|$, $\angle a_k = -\angle a_{-k}$; a_0 is real
2. $x(t) = a_0 + 2 \sum_{k=1}^{\infty} |a_k| \cos\left(\frac{2\pi}{T}kt + \angle a_k\right)$
3. $y(t) = a_0 H_0 + 2 \sum_{k=1}^{\infty} |a_k| |H_k| \cos\left(\frac{2\pi}{T}kt + \angle a_k + \angle H_k\right)$ if $h(t)$ is real

- Exercise: A real-valued signal $x(t)$ has Fourier coefficients: $a_1 = 1 + j$, $a_3 = 2e^{j\frac{\pi}{4}}$. Find a_{-1} and a_{-3} . Show $x(t)$ in real form.

Important Properties of Fourier Series

Given $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$ and $y(t)$ is a modified version of $x(t)$ (doesn't affect T), this table offers a quick way to find the Fourier series coefficients b_k of $y(t)$:

Property	$y(t)$	b_k	Implication
Time shift	$x(t - t_0)$	$e^{-j\frac{2\pi}{T}kt_0} a_k$	Doesn't change $ a_k $ vs. $k\omega$
Time reversal	$x(-t)$	a_{-k}	$a_k^* = a_k = a_{-k}$ if $x(t)$ is real & even
Differentiation	$\frac{d}{dt}x(t)$	$(j\frac{2\pi}{T}k) a_k$	Simplifies calculation

Exercise: Find b_k for the following signals:

- $y(t) = 2x(t) - 3$
- $y(t) = x(t) + x(t - 2)$

注意分开两个信号，am 与 bn，不要二者都使用 k 的序号

Power Distribution Over Frequencies

- **Parseval's power relation:**

$$P_{x(t)} = \frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

- Properties of $x_k(t) = e^{j\frac{2\pi}{T}kt}$

1. $x_k(t) = x_k(t + T)$
2. $\frac{1}{T} \int_T x_k(t) dt = \begin{cases} 0, & k \neq 0 \\ 1, & k = 0 \end{cases} \Rightarrow \frac{1}{T} \int_T x_m(t) x_n^*(t) dt = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases}$

- Power of component $a_k x_k(t)$ is $|a_k|^2$
 - Plot of $|a_k|^2$ vs. $k\omega$ is called the **power line spectrum**
 - Plot of $|a_k|$ vs. $k\omega$ is called the **magnitude line spectrum**
 - Plot of $\angle a_k$ vs. $k\omega$ is called the **phase line spectrum**

$$p(t) = \sum_{m=-\infty}^{\infty} 1 \cdot e^{j2\pi mt}$$

区分冲激串

Important Properties of Fourier Series

Given $x(t)$ with FS coefficients a_k and $y(t)$ with b_k , $z(t)$ involves $x(t)$ and $y(t)$, this table offers a quick way to find the FS coefficients c_k of $z(t)$:

Property	$z(t)$	c_k
Linearity	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Multiplication	$x(t)y(t)$	$\sum_{m=-\infty}^{\infty} a_m b_{k-m}$
Convolution	$\frac{1}{T} \int_T x(\tau) y(t - \tau) d\tau$	$a_k b_k$

信号性质	系数条件
real-valued	$a_{-k} = a_k^*$
even	$a_{-k} = a_k$
odd	$a_{-k} = -a_k$, 且通常 $a_0 = 0$
real and even	$a_k \in \mathbb{R}, a_{-k} = a_k$
real and odd	a_k purely imaginary, $a_{-k} = -a_k$

电感 Inductor

$$v_L(t) = L \frac{di_L(t)}{dt}$$

电容 Capacitor

$$i_C(t) = C \frac{dv_C(t)}{dt}$$